

EC123ATC — Formula Sheet

Principles of Electronics Engineering (Units I, II, IV — from your tutorial material)

UNIT I — Diodes, Rectifiers, Amplifiers

Diode Equation

$$I_D = I_s (e^{V_D/\eta V_T} - 1) \quad V_T = \frac{kT}{q} \quad (\approx 26 \text{ mV at } 300\text{K})$$

- $\eta = 1$ (Ge), $\eta = 2$ (Si)

Bridge Rectifier (Full-Wave)

Quantity	Formula
DC Load Current	$I_{dc} = \frac{2I_m}{\pi}$
DC Load Voltage	$V_{dc} = \frac{2V_m}{\pi} \left(1 + \frac{2R_f}{R_L}\right)$
RMS Load Current	$I_{rms} = \frac{I_m}{\sqrt{2}}$
RMS Load Voltage	$V_{rms} = \frac{V_m}{\sqrt{2}} \left(1 + \frac{2R_f}{R_L}\right)$
Regulation	$\frac{2R_f}{R_L}$
Efficiency	$\eta = \frac{0.812}{1 + 2R_f/R_L}$
Ripple Factor	0.483
PIV	V_m

Bridge Rectifier with Capacitor Filter

$$\gamma = \frac{1}{4\sqrt{3}fCR_L} \quad V_{dc} = \frac{V_m}{1 + \frac{1}{4fCR_L}} \quad \text{Load regulation} = \frac{1}{4fCR_L}$$

Zener Diode

$$R_Z = \frac{\Delta V}{\Delta I} \quad I = I_L + I_Z$$
$$\left[\frac{V_{in(min)} - V_Z}{R} \right] - I_{L(max)} > I_{Z(min)} \quad \left[\frac{V_{in(max)} - V_Z}{R} \right] - I_{L(min)} < I_{Z(max)}, \quad I_{Z(max)} = \frac{P_D}{V_Z}$$

Amplifier Gain & Decibels

$$A_V = \frac{V_o}{V_i} \quad A_I = \frac{I_o}{I_i} \quad A_P = \frac{P_o}{P_i}$$
$$(A_V)_{dB} = 20 \log \left(\frac{V_o}{V_i} \right) \quad (A_I)_{dB} = 20 \log \left(\frac{I_o}{I_i} \right) \quad (A_P)_{dB} = 10 \log \left(\frac{P_o}{P_i} \right)$$

CMRR

$$CMRR = \frac{A_d}{A_c} \quad CMRR(dB) = 20 \log \left| \frac{A_d}{A_c} \right|$$

Cascaded Amplifiers

$$A_V(\text{overall}) = A_{V1} \times A_{V2} \times \dots \times A_{Vn}$$
$$(A_V)_{dB} = (A_{V1})_{dB} + (A_{V2})_{dB} + \dots + (A_{Vn})_{dB}$$

BJT (CE mode)

$$I_C = \beta I_B + I_{CO}(1 + \beta) \quad \beta = \frac{\alpha}{1 - \alpha}$$

DC Load Line: $V_{CC} = I_C R_C + V_{CE} \Rightarrow I_C = \frac{-1}{R_C} V_{CE} + \frac{V_{CC}}{R_C}$ (Saturation point: $I_C = V_{CC}/R_C$ at $V_{CE} = 0$;
Cutoff point: $V_{CE} = V_{CC}$ at $I_C = 0$)

Bandwidth

$$BW = f_H - f_L$$

UNIT II — Feedback, Oscillators, Op-Amps

Negative Feedback

$$A_f = \frac{A}{1 + \beta A} \quad \left| \frac{dA_f}{A_f} \right| = \frac{1}{|1 + \beta A|} \left| \frac{dA}{A} \right|$$

Benefits — all scale by factor $(1 + \beta A)$: input impedance $\uparrow$, output impedance $\downarrow$, bandwidth $\uparrow$, distortion $\downarrow$, noise $\downarrow$, gain stability $\uparrow$.

Barkhausen Criterion

$$|\beta A| = 1 \quad \angle \beta A = 2\pi n, \quad n = 0, 1, 2, \dots$$

RC Phase Shift Oscillator

$$f_0 = \frac{1}{2\pi RC\sqrt{6}} \quad \left(\text{with loading: } f_0 = \frac{1}{2\pi RC\sqrt{6 + 4K}} \right)$$

Gain condition: amplifier gain $A \geq 29$ (to overcome network attenuation).

Wien Bridge Oscillator

$$f = \frac{1}{2\pi\sqrt{R_1 C_1 R_2 C_2}} \quad \text{if } R_1 = R_2 = R, \quad C_1 = C_2 = C : \quad f_0 = \frac{1}{2\pi RC}$$

Op-Amp Circuits

Circuit	Gain / Output
Inverting Amplifier	$A_V = \frac{V_o}{V_i} = -\frac{R_f}{R_1}$
Non-Inverting Amplifier	$A_V = \frac{V_o}{V_i} = 1 + \frac{R_f}{R_1}$
Voltage Follower	$A_V = 1$
Summer (unity R's)	$V_o = -\left[\frac{R_f}{R_1} V_1 + \frac{R_f}{R_2} V_2 + \frac{R_f}{R_3} V_3 \right]$; if $R_f = R_1 = R_2 = R_3$: $V_o = -(V_1 + V_2 + V_3)$
Integrator	$V_o = \frac{-1}{RC} \int_0^t V_i dt$
Differentiator	$V_o = -RC \frac{dV_i}{dt}$

Slew Rate

$$SR = \frac{dV_o}{dt} \text{ (V/}\mu\text{s)} \quad SR = f_{max} \cdot 2\pi \cdot V_m$$

UNIT IV — Communication Systems

Antenna / Wavelength

$$\text{Wavelength} = \frac{\text{Velocity}}{\text{Frequency}} = \frac{3 \times 10^8}{f} \text{ m}$$

Bandwidth

$$BW = f_2 - f_1$$

Amplitude Modulation

$$m = \frac{A_m}{A_c} \quad (0 \leq m \leq 1)$$

$$S_{AM}(t) = (A_c + A_m \sin \omega_m t) \sin \omega_c t$$

Expanded form:

$$S_{AM}(t) = \underbrace{A_c \sin \omega_c t}_{\text{Carrier}} + \underbrace{\frac{mA_c}{2} \cos(\omega_c - \omega_m)t}_{\text{LSB}} - \underbrace{\frac{mA_c}{2} \cos(\omega_c + \omega_m)t}_{\text{USB}}$$

Quantity	Formula
Sideband amplitude	$\frac{mA_c}{2}$
Sideband frequencies	$f_c \pm f_m$
AM bandwidth	$2f_m$
Total transmitted power	$P_T = P_c \left(1 + \frac{m^2}{2}\right)$
Total voltage/current	$V_T = V_c \sqrt{1 + \frac{m^2}{2}}, \quad I_T = I_c \sqrt{1 + \frac{m^2}{2}}$
Effective modulation index (multi-tone)	$m_{eff} = \sqrt{m_1^2 + m_2^2 + m_3^2 + \dots}$ (must not exceed 1)

Quick facts: At $m = 1$ (100% modulation) $\rightarrow$ $P_T = 1.5 P_c$, but only $1/3$ of P_T is in the sidebands (i.e., useful/information power).

Superheterodyne Receiver / Heterodyning

- Standard AM intermediate frequency (IF): **455 kHz**
- Heterodyning combines two frequencies $f_1, f_2 \rightarrow$ sum $(f_1 + f_2)$ filtered out, difference $(f_1 - f_2)$ retained as IF.

Quick-Reference: Common Constants

Symbol	Value
k (Boltzmann constant)	$1.38 \times 10^{-23} \text{ J/K}$ (or $8.617 \times 10^{-5} \text{ eV/K}$)
q (electron charge)	$1.6 \times 10^{-19} \text{ C}$
V_T at 300 K	$\approx 25.85 \text{ mV}$ (commonly rounded to 26 mV)
Speed of light / EM wave velocity	$3 \times 10^8 \text{ m/s}$
Standard AM IF	455 kHz

Note: Unit III (Digital Logic — Boolean algebra, K-maps, adders) has no numerical “formulas” in the same sense; it’s rule/theorem-based (De Morgan’s laws, SOP/POS forms) rather than plug-in equations. Unit V (Embedded Systems) is conceptual with no standard formulas in your material. Let me know if you’d like a rules/theorems reference sheet for Unit III instead.